

StaffDatabaseTest.java – Explanation (Tanglish Version)

■ ****Overview:****

Intha test class `StaffDatabaseTest` nu per iruku. Ithu Spring Boot JUnit test case for verifying ****Staff table la data fetch aagudha illa**** nu check pannrathu. Basically, ithu database la iruka staff records correct ah varudhaa illa nu confirm pannum.

■ ****Code Snippet:****

```
@SpringBootTest
class StaffDatabaseTest {

    @Autowired
    private StaffRepository staffRepository;

    @Test
    void testReadAllStaffFromDatabase() {
        List<Staff> allStaff = staffRepository.findAll();
        assertNotNull(allStaff, "Staff list should not be null");

        if (allStaff.isEmpty()) {
            System.out.println(" No staff found in the database. Please add record");
        } else {
            System.out.println(" Found " + allStaff.size() + " staff record(s) in database");
            allStaff.forEach(staff -> {
                System.out.println("ID: " + staff.getId() +
                    ", Name: " + staff.getName() +
                    ", Email: " + staff.getEmail() +
                    ", Status: " + staff.getStatus());
                assertNotNull(staff.getName(), "Staff name should not be null");
                assertNotNull(staff.getEmail(), "Staff email should not be null");
            });
        }
    }
}
```

■ ****Detailed Explanation (Line by Line):****

- 1■■■ ****@SpringBootTest**** – Ithu Spring Boot environment la test run aaganum nu solrathu.
- 2■■■ ****@Autowired StaffRepository**** – Repository autowire pannirukom; ithula DB connection automatically inject aagum.
- 3■■■ ****testReadAllStaffFromDatabase()**** – Intha method DB la iruka staff list ah fetch pannum.
- 4■■■ ****findAll()**** – StaffRepository la iruka Spring Data JPA method; ithu all staff records ah return pannum.
- 5■■■ ****assertNotNull(allStaff)**** – Data null illa nu confirm pannum.
- 6■■■ ****if (allStaff.isEmpty())**** – Database empty na message print pannum.
- 7■■■ ****else block**** – Records irundha count print pannum, details display pannum (ID, Name, Email, Status).
- 8■■■ ****assertNotNull(staff.getName()) & assertNotNull(staff.getEmail())**** – Each staff la name & email null illa nu verify pannum.

■ ****Summary:****

So, ithu basically one verification test – DB la staff details properly store aagudha, null fields illa nu check pannum. Staff API la data insert panniruka illa na, ithu 'No staff found in the database' nu print pannum.